

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad k=4 \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

(a)

$$df_M = 577 - 4 = 573, N_F = 1000 - 577 = 423, df_F = 423 - 4 = 419$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} = \frac{97161.9174}{573} = 169.55 > F_{0.025, 573, 419} \approx 1$$

$$\Rightarrow F_{0.975, 573, 419} = 0.836 < F^* = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.55}{144.576} = 1.173 < F_{0.025, 573, 419} = 1.194$$

$$\Rightarrow \text{Do not reject } H_0: \sigma_M^2 = \sigma_F^2$$

- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | \mathbf{x}_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | \mathbf{x}_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

$$N_{single} = 400, SSE_{single} = 56231.0382, N_{married} = 600, SSE = 100703.0471$$

$$\hat{\sigma}_{single}^2 = \frac{56231.0382}{400 - 5} = 142.36, \hat{\sigma}_{married}^2 = \frac{100703.0471}{595} = 169.25$$

$$\Rightarrow F^* = \frac{\hat{\sigma}_{married}^2}{\hat{\sigma}_{single}^2} = \frac{169.25}{142.36} = 1.189 > F_{0.05, 595, 395} \approx 1.165$$

$$\Rightarrow \text{Reject } H_0: \hat{\sigma}_{married}^2 = \hat{\sigma}_{single}^2$$

- c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

$$NR^2 = 59.03, S = 5, df = 5 - 1 = 4$$

$$\sigma_i^2 = h(\alpha_1 + \alpha_2 EDUC + \alpha_3 EXPER + \alpha_4 METRO + \alpha_5 FEMALE)$$

$$H_0: \alpha_2 = \alpha_3 = \dots = \alpha_5 = 0, H_1: \text{at least 1 coefficient is not 0}$$

$$\because NR^2 = 59.03 > \chi^2_{4, 0.05} = 9.488$$

$\therefore$ Reject H_0 : No heteroskedasticity

• Does this answer (b)?

No, the BP test only tell us that heterogeneous variance exists, but it cannot determine which variable is causing it.

- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

original variables :	EDUC, EXPER, METRO, FEMALE	4
squares terms :	EDUC ² , EXPER ² , METRO ² , FEMALE ²	2
product terms :	EDUC · EXPER, EDUC · METRO, ...	6

dummy variables $\Rightarrow x^2 = x$

$$\Rightarrow df = 4 + 2 + 6 = 12$$

$$H_0: \alpha_2 = \alpha_3 = \dots = \alpha_5 = 0, H_1: \text{at least 1 coefficient is not 0}$$

$$\because NR^2 = 78.82 > \chi^2_{12, 0.05} = 21.026$$

$\therefore$ Reject H_0 : No heteroskedasticity. The White test provides strong evidence that there's heteroskedasticity in the model.

- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)
	wider	wider	narrower	narrower	narrower

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

No inconsistency. Heteroskedasticity-robust standard errors can be either larger or smaller than OLS se. OLS standard errors are simply biased and inconsistent when heteroskedasticity is present; they don't uniformly overstate or understate the true variance.

- f. If we add *MARRIED* to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

Part (f) examines whether the variable *MARRIED* can explain the difference in average salary. A t -value ≈ 1.0 indicates that *MARRIED* has no significant impact on salary levels.

These two points are not contradictory:

The variance is different for married versus unmarried individuals (conclusion from Part b).

There is no significant difference in average salary between married and unmarried individuals (conclusion from Part f).

A variable can affect the variance of the error, but it does not affect the conditional expected value; the two are independent issues.